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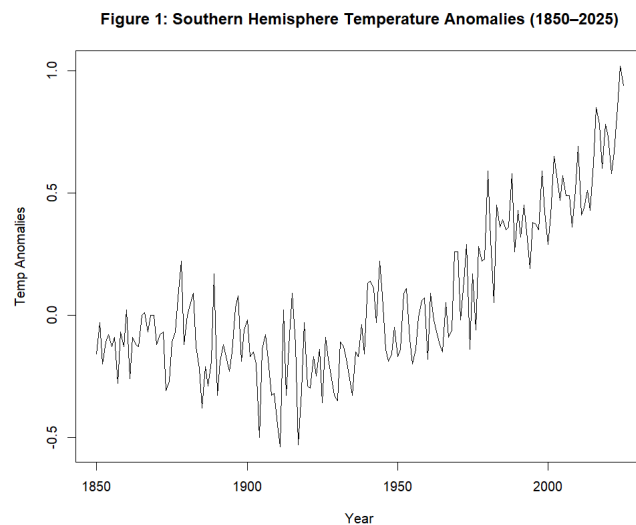
STAT 4363: Applied Time Series Analysis

5 May 2025

Modeling Southern Hemisphere Climate Data: 1850 to 2025

Introduction

The dataset we have chosen to analyze compiles the yearly temperature anomalies of the Southern Hemisphere from 1850 to present. A temperature anomaly is defined as the difference between the observed temperature and a set average or baseline temperature. This dataset, provided by the National Oceanic and Atmospheric Administration, records temperature anomalies with respect to the 1901-2000 temperature average. Additionally, the hemispheric temperature anomaly data comes from NOAA's *Global Surface Temperature Analysis* (*NOAA GlobalTemp*), which combines the comprehensive data collections of both land and ocean surface temperatures. The time series plot is shown in *Figure 1*.

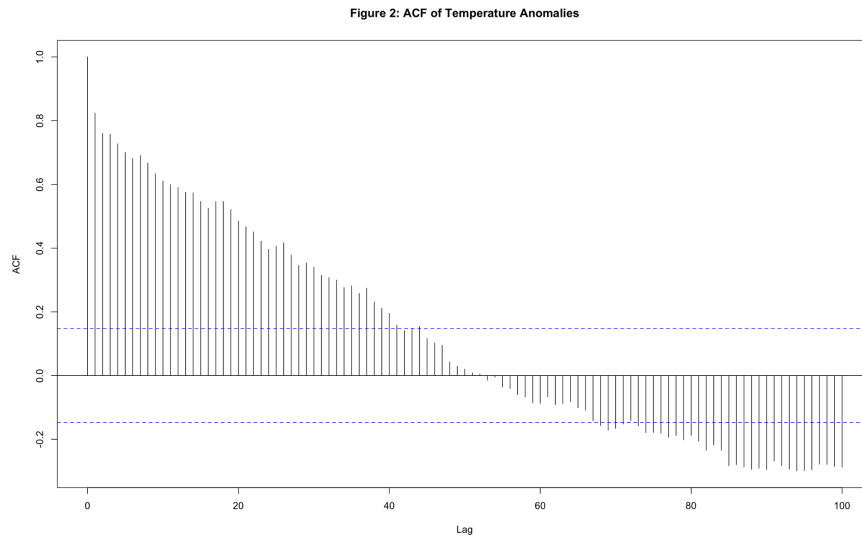


Because temperature anomalies show how much each place is changing relative to its own normal state, analyzing this data can help detect climate change trends on a global or hemispheric scale, highlighting long-term shifts in temperature. Additionally, understanding patterns in temperature anomalies can support global economies in predicting and planning for future impacts.

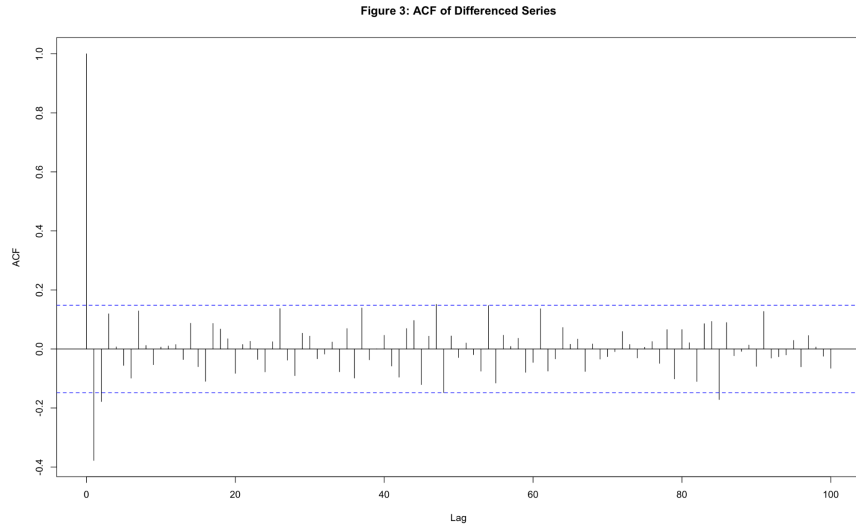
As climate change becomes an increasingly important political and economic issue, it is essential to understand the abnormalities and trends in our changing climate. This understanding helps predict future anomalies and assess the significance of ongoing temperature shifts.

Statistical Analysis

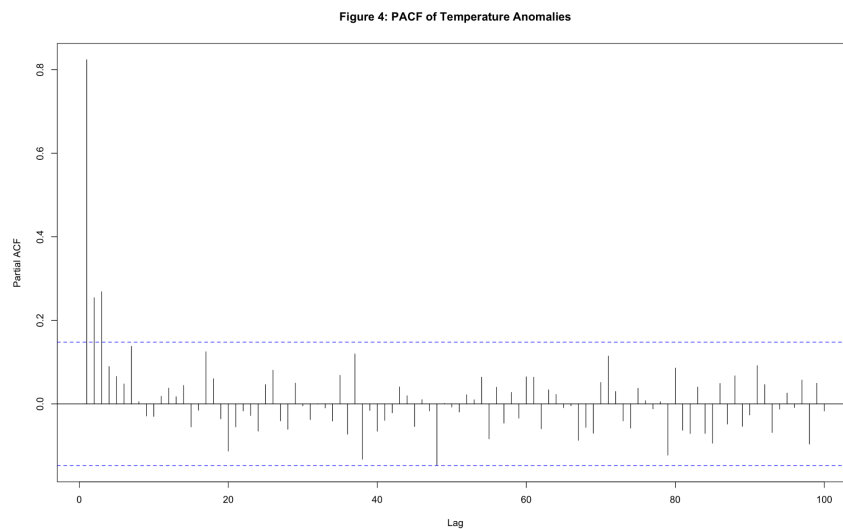
To understand the structure of this dataset, the first visual representation that was applied was an autocorrelation function (ACF) plot, which is useful for detecting non-stationarity, leading data transformations, and choosing the correct ARIMA model. As seen in *Figure 2*, the ACF shows a significant spike at lag 1, but followed by a slowly trailing decay across the lags. This trend suggests that the series is non-stationary due to the upward trend in temperature anomalies over a period of time.



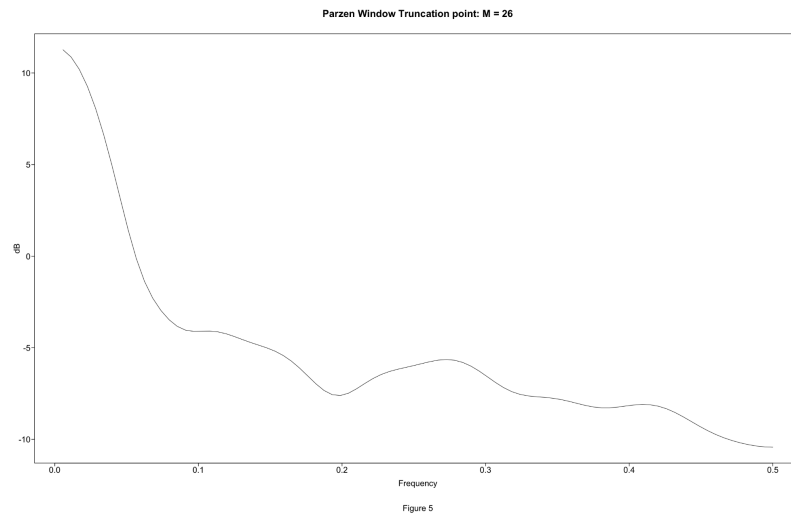
The decay in the autocorrelation suggests that the observations were highly correlated across a long time span, which violates the constant mean requirement for stationarity. Therefore, applying a first-order differencing transformation was ideal for removing the trend and stabilizing the mean. As seen in *Figure 3*, after differencing the ACF the high spike at lag 1 still remained, but there was a much faster decay which confirmed that the data had become stationary. Differencing the ACF was crucial as it made it appropriate for ARIMA modeling, which from the plot supports an MA(1) or MA(2).



To analyze the structure of the temperature anomaly series, a partial autocorrelation function (PACF) plot was examined. A PACF is useful for identifying autoregressive components while modeling. As seen in *Figure 4*, the PACF has a significant spike at lag 1, which is not within the confidence bounds. Following lag 1 are smaller spikes that were within the bounds. Therefore, most of the autocorrelations are depicted by the first lag, as no other higher-order AR components are present. Additionally, the PACF confirms that differencing the model was effective as the series resulted with a low-order ARIMA structure.



The next visual representation used was a spectral density plot which was produced using the parzen window method. The spectral density plot analyzes the frequency content in the dataset. As seen in *Figure 5*, the PACF plot has a higher spectral power at a low frequency of around 0.3 db, after the plot decreases the frequency increases. Therefore, the PACF has low-frequency components. Additionally, there are no peaks or any other higher frequencies, suggesting that there is no periodic pattern. Instead there is a smooth decay which supports the idea that the series is influenced by a long-term process. Therefore, the PACF aligns with what was found earlier in the ACF which also included non-seasonal, trending processes.

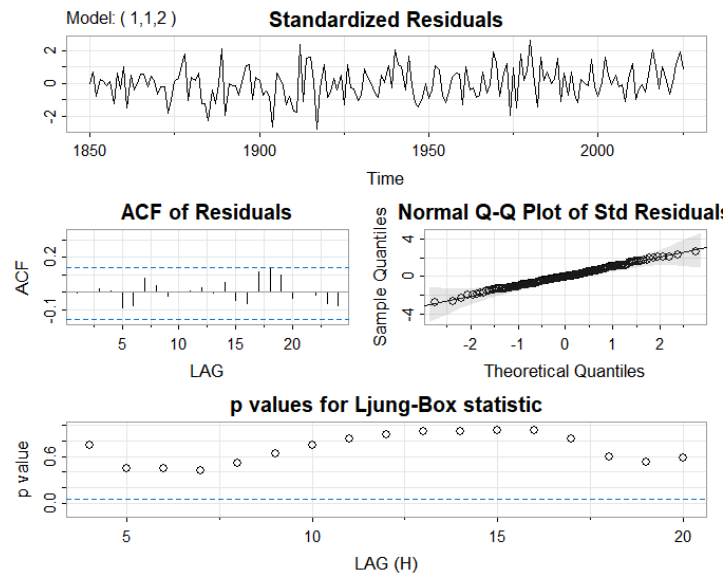


To model the annual temperature anomalies in the Southern Hemisphere from 1850 to 2025, we first visually inspected the time series, which displayed an upward trend over time. This indicated that the series was non-stationary. We confirmed this using the autocorrelation function (ACF), which exhibited slow decay, suggesting the need for differencing. The trend present in the time series, along with the gradual tailing off observed in both the PACF and ACF plots, indicates the presence of autoregressive and moving average components, supporting the use of an ARIMA model as the most appropriate approach for describing the series. Taking the

first difference of the model leaves us with a stationary time series that can be used for determining the ARMA components of the model. The function used to compute this is the *aic5.wge* function from the *tswge* package. This function identified ARIMA(5, 1, 4) as the most accurate model, however the second best was the ARIMA(1, 1, 2) which we ultimately selected due to it being more parsimonious and almost identically accurate. Additionally, the ARIMA(1, 1, 2) performed well and passed residual diagnostic checks, including the ACF of residuals and the Ljung-Box test, as seen in *Figure 6*, indicating that it is one of the models that can accurately describe the time series. The model equation is seen below with AR coefficient -0.5345 and MA coefficients -0.1114 and -0.5430 .

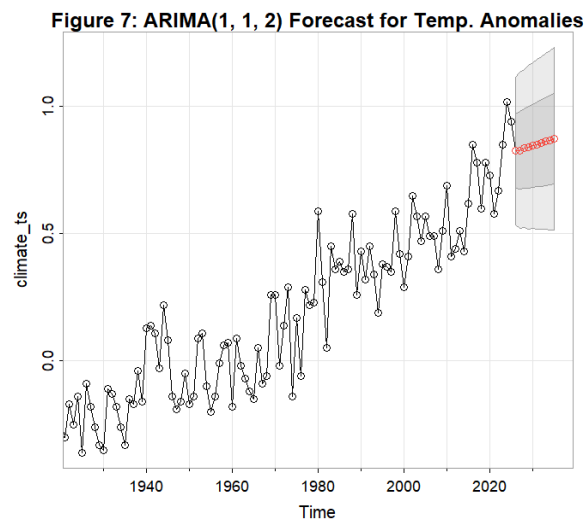
$$\Delta Y_t = 0.0055 - 0.5345\Delta Y_{t-1} - 0.1114a_{t-1} - 0.5430a_{t-2} + a_t$$

Figure 6: ARIMA(1,1,2) Model Fit for Temperature Anomalies



After fitting the ARIMA(1, 1, 2) we used the model to generate a 10-year forecast of Southern Hemisphere temperature anomalies beyond 2025. The resulting forecast shows a continued upward trend which is consistent with the pattern visually identified in the time series.

The forecasted time series plot can be seen in *Figure 7*. While the forecasts appear relatively linear, this is consistent with ARIMA behavior considering the absence of strong seasonal or cyclical components. This forecast also lends itself to a more general question about predicting temperature anomalies in the future. We can reasonably expect that, if current trends persist, temperature anomalies will continue to increase over the coming decade. The steady upward trajectory reflected in the forecast aligns with historical patterns of warming, suggesting that future years are likely to experience above-average deviations from long-term temperature norms.



Conclusion

The dataset analyzed within this study included the annual global temperature anomalies from 1850 to 2025, which captures the long-term changes in the climate. Given the importance of tracking global warming trends, the dataset from *NOAAGlobalTemp* provided us with valuable insights in detecting structural changes and making informed decisions about future climate behavior. The time series, ACF, PACF, spectral density estimation, and ARIMA modeling all resulted in an upward trend and having non-stationarity, which was confirmed by the ACF trailing decay. Additionally, a first order differencing transformation was applied in order to

make the data stationary. The spectral analysis reinforced these findings from the PACF as well, as there were low-frequency components that were present throughout the data.

To model this behavior, an ARIMA(1,1,2) was chosen as it was well-balanced and could accurately capture the structure of the series. Therefore, the resulting model was used to predict the temperature anomalies for the next ten years, which generated an upward trend. This outcome supports our conclusion, as the absence of strong seasonal or cyclical components suggests that the temperature anomalies are likely to continue to increase- further reinforcing the presence of long-term climate change. Therefore, the analysis was successful in understanding the historical patterns and future trajectory of temperature anomalies in the Southern Hemisphere.

Reference

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